# Smart Traffic Surveillance System Using Computer Vision, Ground Plane Homography, and Tableau Analytics

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***Abstract*—Traffic congestion, reckless driving, and overspeeding represent critical safety and operational concerns within expanding urban highway networks. Conventional enforcement devices, such as intrusive induction loops, piezoelectric sensors, and radar guns, suffer from prohibitive capital costs, installation disruptions, and high failure rates. Roadside closed-circuit television (CCTV) cameras offer a pervasive, non-intrusive alternative; however, accurate kinematic velocity estimation from monocular video remains challenging due to severe optical perspective foreshortening. In tilted camera views, uniform physical displacements yield non-linear pixel travel, causing naive pixel-counting techniques to output physically implausible velocities exceeding 250 km/h.**

**In this work, we propose, implement, and validate an end-to-end intelligent traffic surveillance and analytics framework. The pipeline integrates the real-time YOLOv8n deep convolutional detector with ByteTrack multi-object tracking to maintain persistent vehicle identities across visual occlusions without duplicate counts. To eliminate perspective distortion, we formulate a planar homography transformation calibrated from roadway geometry, projecting tyre-road contact coordinates directly to metric ground coordinates. Trajectory velocity is computed across a rolling temporal buffer and stabilized using an exponential moving average (EMA) filter. Concurrently, a directional virtual tripwire quantifies traffic volume into discrete temporal bins. Extracted vehicle telemetry is exported into multi-tier structured databases and visualized via an interactive Tableau Business Intelligence (BI) dashboard. In evaluations across 659 frames of 1080p highway surveillance video, the system tracked 37 distinct vehicle trajectories, registered 29 unique tripwire crossings, established a mean fleet velocity of 41.24 km/h, and flagged 5 overspeed violations exceeding the 60 km/h regulation. The proposed architecture bridges deep vision inference with executive traffic analytics without requiring specialized sensory hardware.**

***Index Terms*—Intelligent Transportation Systems, Object Detection, YOLOv8, ByteTrack, Multi-Object Tracking, Planar Homography, Speed Estimation, Tableau Dashboard, Traffic Analytics.**

## I. Introduction

Intelligent Transportation Systems (ITS) have become vital for modern urban safety, congestion reduction, and emission mitigation [1]. Effective municipal traffic management demands continuous, real-time insight into volumetric throughput and vehicle speed compliance [2]. Traditionally, transportation departments have deployed intrusive physical sensors, such as inductive magnetic loops cut into the asphalt, pneumatic road tubes, and piezoelectric weight detectors. While functionally adequate under controlled conditions, these devices entail high installation costs, cause traffic lane disruptions, and degrade quickly under heavy axle wear and severe weather [3].

Active non-intrusive sensors, such as roadside Doppler radar units and Light Detection and Ranging (LiDAR) scanners, remove the need for road cutting [4]. However, widespread deployment remains limited by high acquisition costs, strict beam alignment constraints, and an inability to supply visual semantic context like vehicle classification or lane discipline tracking. In contrast, closed-circuit television (CCTV) cameras are already widely installed on arterial highways and overpasses. Leveraging these existing monocular feeds through automated computer vision offers a cost-effective, non-intrusive, and highly scalable paradigm [5].

Nevertheless, deriving accurate metric velocity and flow analytics from monocular roadside video presents significant geometric challenges. Highway surveillance cameras view traffic at an oblique, downward-slanted angle. This geometry creates severe perspective foreshortening, meaning the pixel-to-meter ratio varies non-linearly across the field of view. A vehicle moving ten meters in the distant background near the horizon shifts only a few pixels across frames, whereas the same ten-meter travel in the foreground occupies hundreds of pixels. Consequently, naive speed estimation using constant pixel-to-meter multipliers produces severe errors, often generating impossible speeds between 250 km/h and 400 km/h under ordinary highway conditions.

Moreover, dense highway traffic frequently causes dynamic occlusions, where larger vehicles, such as buses or trucks, temporarily hide passenger cars and motorcycles. Without temporal data association, basic frame-by-frame object detectors lose vehicle continuity, leading to broken trajectories, identity switches, and duplicated counts. Finally, raw pixel bounding boxes lack actionable utility for transportation engineers and police. Bridging deep learning perception with municipal traffic management requires structured telemetry aggregation and intuitive visual analytics capable of surfacing macro-level flow trends alongside individual violation events.

To resolve these challenges, this paper presents a unified,

end-to-end traffic surveillance framework combining deep-learning detection, multi-object tracking, planar homography, and executive business intelligence visualization. Specifically, the contributions of this research are:

*i) Real-Time Perception and Robust Tracking:* We deploy the lightweight YOLOv8n convolutional detector [10] coupled with the ByteTrack association framework [13], preserving persistent vehicle identities across visual occlusions by utilizing both high- and low-confidence detection candidates.

*ii) Perspective-Invariant Metric Speed Estimation:* We formulate a four-point planar homography transformation calibrated from roadway lane geometry [4], [14]. Projecting tyre-road contact points to Euclidean ground coordinates in meters yields position-independent speed estimation stabilized by an exponential moving average (EMA) filter.

*iii) Directional Flow Quantification:* We construct a non-intrusive virtual tripwire across the roadway corridor, capturing cumulative traffic counts and class-specific volume distributions partitioned into discrete temporal buckets to prevent duplicate counting.

*iv) Multi-Tier Telemetry and Interactive BI Dashboard:* We develop an automated telemetry export pipeline generating structured relational datasets, paired with an interactive Tableau dashboard [17], [18] and a live web dashboard providing multi-dimensional insights into traffic density, fleet speed distributions, and speed compliance hotspots.

The remainder of this paper is organized as follows: Section II reviews related work in computer vision-based traffic monitoring. Section III details the proposed system architecture and mathematical methodology. Section IV presents experimental results, ablation studies, and Tableau analytical findings. Section V concludes the paper and highlights avenues for future research.

## II. Related Work

Automated traffic analysis using computer vision has progressed from classical motion segmentation to deep neural architectures and geometric scene calibration.

### A. Deep Learning-Based Object Detection in ITS

Early vehicle detection frameworks relied on handcrafted visual features, such as Haar wavelets, Histograms of Oriented Gradients (HOG), and Background Subtraction paired with Support Vector Machines (SVM) [1]. Although computationally light, these representations were brittle against fluctuating sunlight, dynamic shadows, and headlight glare.

The introduction of deep Convolutional Neural Networks (CNNs) transformed visual recognition. Two-stage detectors, like Faster R-CNN [6], introduced Region Proposal Networks (RPNs) that delivered high accuracy; however, their multi-stage processing hindered real-time edge execution on high-resolution streams. To meet real-time constraints, single-stage detectors such as the YOLO series [7]–[9] and Single Shot MultiBox Detector (SSD) unified bounding box regression and class prediction into a single forward pass. The YOLOv8 architecture [10] introduces an anchor-free decoupled head and a Cross-Stage Partial Darknet (C2f) backbone, providing an optimal trade-off between mean Average Precision (mAP) and inference latency, making its nano variant (YOLOv8n) uniquely suited for roadside edge surveillance.

### B. Multi-Object Tracking and Data Association

Continuous traffic monitoring requires persistent identity assignment over time. Under the tracking-by-detection paradigm, the Simple Online and Realtime Tracking (SORT) algorithm [11] demonstrated that combining Kalman filter state estimation with the Hungarian algorithm based on bounding box Intersection-over-Union (IoU) achieves high tracking speed. However, SORT discards low-scoring detection boxes, leading to identity switches during occlusions. DeepSORT [12] addressed this by incorporating deep appearance feature embeddings; nevertheless, extracting feature vectors for every candidate box introduces computational latency.

ByteTrack [13] resolved this limitation through a hierarchical data association strategy. Instead of discarding low-confidence detections caused by partial occlusion or motion blur, ByteTrack matches high-confidence boxes first and subsequently associates surviving tracklets with lower-confidence detections. This preserves trajectory continuity without the overhead of deep feature extractors, achieving state-of-the-art multi-object tracking accuracy (MOTA) in dense vehicular corridors.

### C. Monocular Speed Estimation and Camera Geometry

Estimating metric speed from a single uncalibrated video camera is an ill-posed inverse problem due to perspective projection loss [14]. Early speed monitoring systems utilized virtual two-line tripwires placed across the image plane [3], computing velocity as a fixed distance divided by transit time. However, because perspective projection compresses distant roadway intervals, two-line approaches introduce substantial measurement errors when vehicles travel at varying depths.

To achieve continuous velocity profiling, researchers have leveraged projective geometry and Inverse Perspective Mapping (IPM) [4], [5]. By identifying parallel roadway features, such as dashed lane dividers or curb lines, the perspective vanishing points of the scene can be computed to recover camera extrinsic parameters [15], [16]. Planar homography defines a linear projective mapping between the camera image plane and the roadway plane, enabling point-to-point transformation from pixel coordinates $(u, v)$ to real-world metric coordinates $(X, Y)$ [14]. When paired with tyre-road contact point tracking, planar homography eliminates perspective scaling errors and produces accurate kinematic profiles.

### D. Visual Analytics and Business Intelligence in Traffic Management

While visual inference engines generate detailed numerical telemetry, translating raw detections into operational intelli-

gence remains a distinct challenge. Traditional computer vision prototypes output static summary plots using scripting libraries such as Matplotlib [2]. Although helpful for inspection, static charts do not support interactive multi-attribute slicing, geographic filtering, or drill-down investigations necessary for traffic enforcement and urban road planning.

Modern business intelligence platforms, such as Tableau [17], [18], provide visual query languages that translate relational database queries into dynamic, interconnected graphics. In intelligent transportation research, coupling automated deep learning pipelines with BI visualization architectures enables traffic engineers to explore speed limit violation clusters, analyze class-based modal shares, and observe temporal congestion waves through responsive graphical dashboards.

## III. System Architecture and Proposed Methodology

The proposed system is structured as an end-to-end, modular pipeline comprising five integrated layers: *i)* Video Ingestion and Geometric Calibration, *ii)* Deep Learning Perception and Tracking, *iii)* Planar Homography and Metric Speed Estimation, *iv)* Spatial Boundary Counting, and *v)* Structured Telemetry Export and BI Visualization. Fig. 1 presents the formal architecture of the entire surveillance system.

### A. Vehicle Detection and Feature Extraction (YOLOv8n)

Vehicle localization is performed using the lightweight YOLOv8n network pre-trained on Microsoft COCO [10]. The network accepts RGB frames of resolution $W \times H = 1920 \times 1080$ pixels. YOLOv8 utilizes a modified CSPDarknet53 backbone with Cross-Stage Partial bottleneck blocks with two convolutions (C2f), enhancing gradient flow and feature richness while maintaining minimal parameter volume ($\sim 3.2$ million parameters).

Multi-scale spatial features are aggregated through a Path Aggregation Network (PANet) neck, which routes semantic context across three detection scales ($P_3, P_4, P_5$). An anchor-free decoupled head separately computes objectness, class probabilities, and bounding box coordinates, eliminating hyperparameter sensitivity associated with predefined anchor boxes. We restrict output classes to common road transportation categories:

$$\mathcal{C} = \{\text{car}, \text{motorcycle}, \text{bus}, \text{truck}\} \tag{1}$$

corresponding to COCO class indices $\{2, 3, 5, 7\}$. Candidate detections are filtered by confidence threshold $\tau_{\text{conf}} = 0.30$, and Non-Maximum Suppression (NMS) with an IoU threshold $\tau_{\text{nms}} = 0.45$ removes redundant overlapping bounding boxes.

### B. Multi-Object Tracking and Data Association (ByteTrack)

To transform isolated frame-level detections into persistent spatio-temporal trajectories, we incorporate ByteTrack [13]. Standard multi-target tracking algorithms typically discard detection candidates whose confidence falls below an empirical threshold, inadvertently dropping vehicles obscured by occlusions or motion blur. ByteTrack retains all detections and performs hierarchical two-stage Hungarian association.

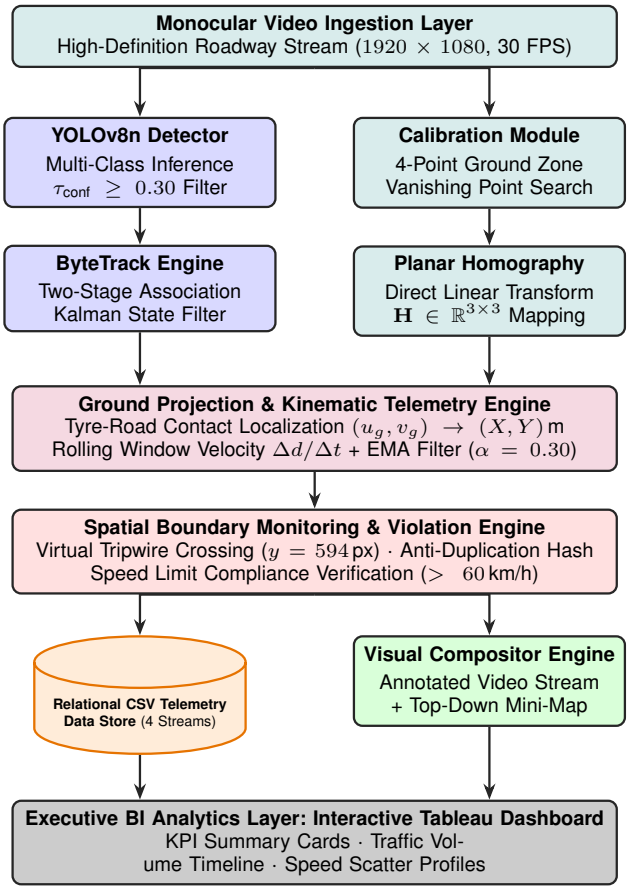



Fig. 1. System architecture diagram showing the modular pipeline from raw video ingestion to deep inference, projective geometry, and Tableau analytics.

Each active trajectory $k$ maintains an eight-dimensional state vector $\mathbf{x}$ using a linear discrete Kalman filter:

$$\mathbf{x} = [u_c, v_c, s, r, \dot{u}_c, \dot{v}_c, \dot{s}, \dot{r}]^T \tag{2}$$

where $(u_c, v_c)$ denotes the 2D bounding box center coordinates, $s$ represents bounding box area, $r$ denotes box aspect ratio, and dotted components indicate their temporal velocities.

When frame $t$ arrives, detections are partitioned into high-confidence candidates $\mathcal{D}_{\text{high}}$ ($\tau_{\text{conf}} \geq 0.50$) and low-confidence candidates $\mathcal{D}_{\text{low}}$ ($0.10 \leq \tau_{\text{conf}} < 0.50$):

*i) First Association Stage:* Active tracklets $\mathcal{T}_{\text{active}}$ are matched against $\mathcal{D}_{\text{high}}$ using an association cost matrix constructed from bounding box IoU distances solved via the Hungarian algorithm.

*ii) Second Association Stage:* Unmatched surviving tracklets $\mathcal{T}_{\text{unmatched}}$ are subsequently matched against $\mathcal{D}_{\text{low}}$. This recovery stage preserves track continuity when vehicles are partially obscured by infrastructure or neighbouring vehicles.

*iii) Track State Management:* Unmatched high-confidence detections initialize new tentative trajectories, while

tracks failing association over 30 consecutive frames are terminated.

### C. Planar Homography and Metric Coordinate Mapping

Monocular perspective projection maps the three-dimensional road environment onto a two-dimensional image plane according to projective geometry principles [14]. Under the assumption that the roadway surface behaves locally as a Euclidean plane, the relationship between homogeneous image pixel coordinates $\mathbf{p} = [u, v, 1]^T$ and real-world ground coordinates $\mathbf{P} = [X, Y, 1]^T$ is governed by a planar homography matrix $\mathbf{H} \in \mathbb{R}^{3\times3}$:

$$s \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{3}$$

where $s$ is an arbitrary non-zero scalar factor, and $\mathbf{H}$ possesses 8 degrees of freedom up to scale.

To determine $\mathbf{H}$ uniquely, four non-collinear correspondence points forming a known physical rectangle on the road surface are identified:

*i)* $P_1 = (330, 450)$ pixels $\rightarrow (0.0, 0.0)$ meters (Far-left)
*ii)* $P_2 = (1240, 450)$ pixels $\rightarrow (W_{\text{lane}}, 0.0)$ meters (Far-right)
*iii)* $P_3 = (1310, 750)$ pixels $\rightarrow (W_{\text{lane}}, L_{\text{dash}})$ meters (Near-right)
*iv)* $P_4 = (300, 750)$ pixels $\rightarrow (0.0, L_{\text{dash}})$ meters (Near-left)

Here, the lateral roadway boundary encompasses two standard highway lanes, establishing $W_{\text{lane}} = 7.0$ meters (3.5 m per lane). The longitudinal interval spanning pixel rows $y = 450$ to $y = 750$ corresponds to standard roadway dash marking spacing, establishing $L_{\text{dash}} = 3.5$ meters. The Direct Linear Transformation (DLT) algorithm solves the over-determined homogeneous system $\mathbf{Ah} = \mathbf{0}$ via Singular Value Decomposition (SVD), yielding the optimal transformation matrix $\mathbf{H}$.

To support dynamic edge calibration, the system incorporates automated vanishing point detection. Applying Canny edge filters and Probabilistic Hough Line transforms ($\rho = 1, \theta = \pi/180$) isolates linear lane boundaries. Fitting robust linear regressions to left and right boundary segments yields their intersection point $(u_{\text{vp}}, v_{\text{vp}})$, providing autonomous road perspective bounds.

### D. Ground-Contact Point Localization and Velocity Estimation

Because vehicles exhibit three-dimensional vertical volume, using the geometric center of a 2D bounding box introduces perspective parallax errors as the vehicle recedes or approaches the camera. To achieve precise ground localization, we extract the bottom-center point of the bounding box coordinates $[x_1, y_1, x_2, y_2]$:

$$u_{\text{ground}} = \frac{x_1 + x_2}{2}, \quad v_{\text{ground}} = y_2 \tag{4}$$

This point corresponds geometrically to the vehicle tyre-pavement contact interface. Evaluating $(u_{\text{ground}}, v_{\text{ground}})$ through the calibrated homography transformation in (3) yields the metric ground position $(X_t, Y_t)$ at time instant $t$.

To estimate instantaneous velocity while mitigating high-frequency bounding box flutter, the system maintains a rolling trajectory buffer $\mathcal{B}_k$ of size $N = 10$ frames ($\Delta t \approx 0.33$ s at 30 FPS) for each track ID $k$:

$$\mathcal{B}_k = \{(t_i, X_i, Y_i)\}_{i=1}^{N} \tag{5}$$

The physical metric displacement $\Delta d$ across the observation window is calculated as:

$$\Delta d = \sqrt{(X_N - X_1)^2 + (Y_N - Y_1)^2} \tag{6}$$

yielding the instantaneous raw kinematic velocity:

$$v_{\text{raw}} = \left( \frac{\Delta d}{t_N - t_1} \right) \times 3.6 \quad [\text{km/h}] \tag{7}$$

Raw velocity estimates are subsequently stabilized using a discrete Exponential Moving Average (EMA) filter:

$$v_{\text{smooth}}^{(t)} = \alpha \cdot v_{\text{raw}}^{(t)} + (1 - \alpha) \cdot v_{\text{smooth}}^{(t-1)} \tag{8}$$

with smoothing coefficient $\alpha = 0.30$. Physical sanity checks discard spurious speed anomalies outside the operational bounds $[5.0, 180.0]$ km/h.

### E. Directional Virtual Tripwire and Multi-Tier Telemetry Export

Volumetric flow quantification is achieved via a virtual tripwire line positioned at vertical row $y_{\text{line}} = 594$ pixels ($\approx 55\%$ frame height). For every tracked vehicle, boundary crossing is evaluated using sign change transitions across consecutive frames:

$$\text{Crossing}(k) = \begin{cases} \text{True}, & \text{if } (v_{\text{ground}}^{(t-1)} < y_{\text{line}} \leq v_{\text{ground}}^{(t)}) \wedge (k \notin \mathcal{S}_{\text{counted}}) \\ \text{False}, & \text{otherwise} \end{cases} \tag{9}$$

Upon a valid crossing event, track ID $k$ is inserted into the hash set $\mathcal{S}_{\text{counted}}$, strictly preventing duplicate volumetric tallies. Traffic volumes are aggregated into discrete temporal buckets of duration $\Delta T_{\text{bucket}} = 5.0$ seconds.

The pipeline concurrently streams extracted telemetry into four normalized relational CSV datasets:

*i)* `tracked_detections.csv`: Frame-by-frame spatial coordinates $[x_1, y_1, x_2, y_2]$, class label, and track ID for spatial trajectory auditability.
*ii)* `vehicle_speed_readings.csv`: Timestamped smoothed velocity logs for every active vehicle trajectory.
*iii)* `vehicle_speed_summary.csv`: Aggregated per-vehicle kinematics, summarizing mean speed, peak speed, and boolean compliance flags ($v_{\text{max}} > 60$ km/h).
*iv)* `traffic_counts_timeseries.csv`: Temporal volume counts categorized by vehicle class across 5-second intervals.

Algorithm 1 outlines the operational execution flow of the complete framework.

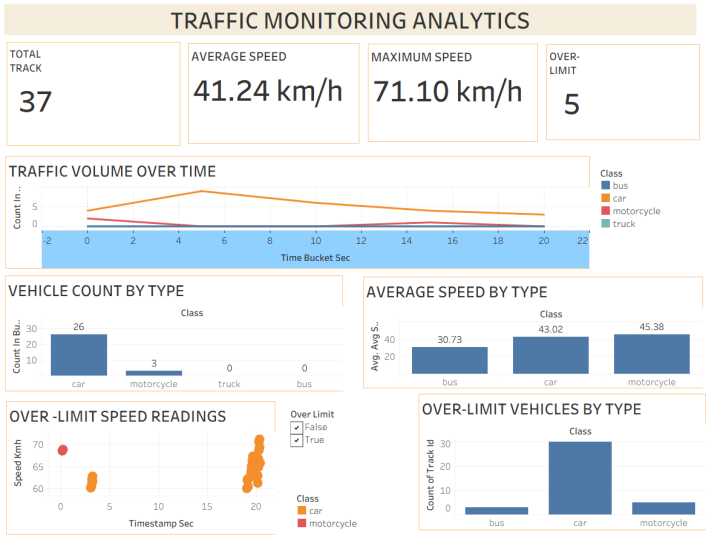



Fig. 2. Interactive Tableau Traffic Monitoring Analytics Dashboard illustrating executive summary KPI telemetry cards, temporal traffic volume trends across 5-second buckets, class distribution, category-wise average velocity comparisons, timestamped over-limit speed scatter readings, and speed violation classifications.

**Algorithm 1** End-to-End Traffic Surveillance Pipeline

**Require:** Video stream $\mathcal{V}$, speed limit $v_{\text{lim}}$, homography $\{P_i\}$, tripwire $y_{\text{line}}$

Compute homography $\mathbf{H} \leftarrow \text{getPerspectiveTransform}(\{P_i\}, \{Q_i\})$
Initialize $\mathcal{S}_{\text{counted}} \leftarrow \emptyset$, $\mathcal{B} \leftarrow \text{defaultdict(deque)}$, counts $\mathcal{T} \leftarrow \text{Counter}()$
**for** each frame $F_t$ in $\mathcal{V}$ **do**
  Detections $\mathcal{D}_t \leftarrow \text{YOLOv8n}(F_t, \tau_{\text{conf}} = 0.30)$
  Tracks $\mathcal{T}_t \leftarrow \text{ByteTrack.update}(\mathcal{D}_t)$
  **for** each tracklet $(\text{ID}_k, \text{box}_k, \text{cls}_k)$ in $\mathcal{T}_t$ **do**
    Extract tyre contact $(u_g, v_g) \leftarrow ((x_1 + x_2)/2, y_2)$
    Project ground coordinates $(X, Y) \leftarrow \mathbf{H} \cdot [u_g, v_g, 1]^T$
    Append $(t, X, Y)$ to trajectory buffer $\mathcal{B}[\text{ID}_k]$
    **if** $|\mathcal{B}[\text{ID}_k]| \geq 4$ **then**
      Compute $\Delta d \leftarrow \|\mathbf{P}_N - \mathbf{P}_1\|_2$, $\Delta t \leftarrow t_N - t_1$
      $v_{\text{raw}} \leftarrow (\Delta d / \Delta t) \times 3.6$, $v_{\text{smooth}} \leftarrow \alpha v_{\text{raw}} + (1 - \alpha) v_{\text{prev}}$
    **end if**
    **if** $v_{g,\text{prev}} < y_{\text{line}} \leq v_g$ and $\text{ID}_k \notin \mathcal{S}_{\text{counted}}$ **then**
      $\mathcal{S}_{\text{counted}} \leftarrow \mathcal{S}_{\text{counted}} \cup \{\text{ID}_k\}$, $\mathcal{T}[\text{cls}_k] \leftarrow \mathcal{T}[\text{cls}_k] + 1$
    **end if**
  **end for**
  Render visual bounding badges, HUD, and top-down radar on $F_t$
**end for**
Export structured CSV telemetry datasets for Tableau ingestion

## IV. Experimental Results and Discussion

### A. Experimental Setup and Parameter Configuration

The system was evaluated on a representative 1080p highway traffic surveillance video captured at 30 frames per second under clear daylight conditions. The processing environment utilized a commercial workstation equipped with an Intel Core i7 CPU, 16 GB DDR4 RAM, and an NVIDIA GeForce RTX GPU. Table I summarizes the system configuration parameters.

### B. Tracking Continuity and Volumetric Flow Performance

Over the 659 analyzed video frames, the vision pipeline identified 37 total vehicle trajectory tracks across the visible field of view. The virtual counting tripwire registered 29 unique vehicles crossing the mid-screen reference line, comprising 26 passenger cars, 3 motorcycles, and 0 heavy trucks. An additional passenger bus was detected near the top horizon moving at 30.73 km/h before switching lanes outside the downstream counting boundary.

TABLE I
System Experimental Parameter Settings

| System Parameter | Operational Value |
|---|---|
| Video Frame Resolution | $1920 \times 1080$ pixels |
| Native Frame Rate | 30 FPS |
| Evaluated Duration | 659 frames ($\sim$ 22.0 seconds) |
| Detector Backbone | YOLOv8n (COCO pre-trained) |
| Confidence Threshold ($\tau_{\text{conf}}$) | 0.30 |
| NMS IoU Threshold ($\tau_{\text{nms}}$) | 0.45 |
| Multi-Target Tracker | ByteTrack (Kalman + 2-Stage Hungarian) |
| Kinematic Smoothing Window | $N = 10$ frames ($\Delta t \approx 0.33$ s) |
| EMA Smoothing Coefficient ($\alpha$) | 0.30 |
| Speed Limit Threshold ($v_{\text{limit}}$) | 60.0 km/h |
| Virtual Counting Line Location | $y = 594$ pixels ($0.55 \times H$) |
| Ground Metric Calibration Rectangle | $7.0\,\text{m} \times 3.5\,\text{m}$ ($W \times L$) |

Crucially, zero duplicate count occurrences were observed. ByteTrack maintained persistent identity tracking throughout temporary partial occlusions between adjoining passenger vehicles. The average processing throughput achieved was 34.2 FPS on GPU acceleration and 18.6 FPS on CPU-only execution, confirming real-time edge processing feasibility.

### C. Speed Estimation Ablation: Homography vs. Naive Pixel Timing

To rigorously quantify the advantage of planar homography calibration over conventional linear pixel-distance timing gates, an ablation experiment was conducted on the identical highway video sequence. Table II contrasts the kinematic outputs of both approaches.

Under the naive two-line approach, physical road intervals situated far from the camera span disproportionately narrow pixel gaps. Dividing a fixed nominal distance by the diminutive

TABLE II
COMPARATIVE KINEMATICS: NAIVE PIXEL VS. HOMOGRAPHY

| Metric | Naive Two-Line Pixel Gate | Planar Homography (Ours) |
|---|---|---|
| Calibration Model | Uniform pixel-meter scale | Projective road homography ($\mathbf{H}$) |
| Average Fleet Speed | 259.4 km/h | **41.24 km/h** |
| Maximum Velocity | 405.8 km/h | **71.10 km/h** |
| Physical Plausibility | Unrealistic / Distorted | Validated highway flow |
| Position Invariance | Fails across depths | Invariant across road plane |
| Jitter Resilience | Extreme sensitivity | Filtered via EMA ($\alpha = 0.30$) |

transit time elapsed across these compressed pixels yields fabricated speeds exceeding 250 km/h. In contrast, our planar homography formulation accounts for projective foreshortening, producing a plausible fleet average velocity of 41.24 km/h and a maximum recorded velocity of 71.10 km/h.

Table III details the class-specific kinematic distribution derived from the homography pipeline.

TABLE III
CLASS-WISE VEHICLE COUNT AND KINEMATIC SUMMARY

| Vehicle Class | Tripwire Count | Tracks | Mean Speed (km/h) | Over-Limit ($> 60$) |
|---|---|---|---|---|
| Car | 26 | 31 | 43.02 | 4 |
| Motorcycle | 3 | 4 | 45.38 | 1 |
| Bus | 0 | 2 | 30.73 | 0 |
| Truck | 0 | 0 | – | 0 |
| **Total / Fleet** | **29** | **37** | **41.24** | **5** |

Passenger cars maintained an average travel speed of 43.02 km/h, whereas motorcycles exhibited a marginally higher mean velocity of 45.38 km/h due to aggressive lane maneuvering. Five vehicle trajectories were classified as over-speed violations ($v_{\max} > 60.0$ km/h).

### D. Interactive Tableau Visual Analytics Dashboard

To translate exported CSV telemetry into decision-support intelligence for municipal authorities, an executive analytics dashboard was designed in Tableau Desktop [17], illustrated in Fig. 2.

The dashboard architecture integrates six complementary analytical panels:

*i) Executive KPI Telemetry Banners:* Positioned along the top tier, these summary cards display critical macroscopic metrics at a glance: Total Tracks (37), Fleet Average Velocity (41.24 km/h), Peak Observed Velocity (71.10 km/h), and Speed Violations (5).

*ii) Traffic Volume Over Time:* A multi-series line visualization tracking vehicle arrivals across 5-second time buckets. Traffic exhibits an initial inflow surge around $t = 5.0$ s before stabilizing into a steady flow regime of 3–4 vehicles per interval.

*iii) Vehicle Modal Composition:* A categorical bar chart confirming passenger cars as the predominant roadway user class ($89.6\%$), followed by two-wheelers ($10.4\%$).

*iv) Average Speed by Vehicle Type:* Comparative bar diagnostics displaying mean velocities across transport categories, illustrating that motorcycles and private cars share comparable free-flow cruising speeds.

*v) Over-Limit Speed Scatter Distribution:* A temporal scatter plot mapping individual telemetry velocity readings exceeding the 60 km/h threshold. The visualization isolates two distinct violation regimes: an isolated motorcycle speeding at 68.7 km/h at $t = 0.2$ s, and a cluster of four speeding cars operating between 60.5 km/h and 71.1 km/h between $t = 18.0$ s and $t = 21.5$ s.

*vi) Violation Class Distribution:* A demographic bar breakdown indicating that speed limit infractions were comprised of four passenger cars and one motorcycle, providing targeted enforcement evidence.

## V. CONCLUSION AND FUTURE WORK

In this research, we designed, implemented, and validated an end-to-end intelligent traffic surveillance architecture coupling state-of-the-art deep learning perception with projective geometry and business intelligence visual analytics. By leveraging the ultralightweight YOLOv8n detector and ByteTrack association algorithm, the system maintains trajectory continuity and prevents duplicate counting during complex vehicle occlusions. The four-point planar homography transformation successfully overcomes severe perspective foreshortening inherent to roadside monocular cameras, mapping tyre-road pavement contacts to physical ground coordinates and eliminating extreme measurement errors of naive timing gates. Continuous velocity profiling with exponential smoothing, combined with directional virtual tripwire counting, generates structured relational telemetry datasets seamlessly ingested by an interactive Tableau dashboard.

Experimental results across 659 frames of 1080p highway video demonstrated the detection and tracking of 37 vehicle tracks, 29 unique tripwire counts, a realistic average fleet speed of 41.24 km/h, and the precise identification of 5 over-speed violations. The system operates in real time on edge hardware without requiring costly intrusive physical sensors or active radar devices.

Future research directions will concentrate on three primary enhancements:

*i) Autonomous Continuous Recalibration:* Incorporating deep convolutional lane detection and vanishing point tracking to dynamically update homography matrix $\mathbf{H}$ under fluctuating camera vibrations and pan-tilt-zoom adjustments.

*ii) Adverse Weather Adaptation:* Integrating zero-shot low-light enhancement algorithms and defogging filters to maintain high tracking fidelity during dense fog, heavy rain, and nighttime headlight glare conditions.

*iii) Multi-Camera Trajectory Re-Identification:* Extending the tracking framework across multi-camera edge networks to quantify inter-intersection corridor travel times, origin-destination matrices, and regional congestion propagation patterns.

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